

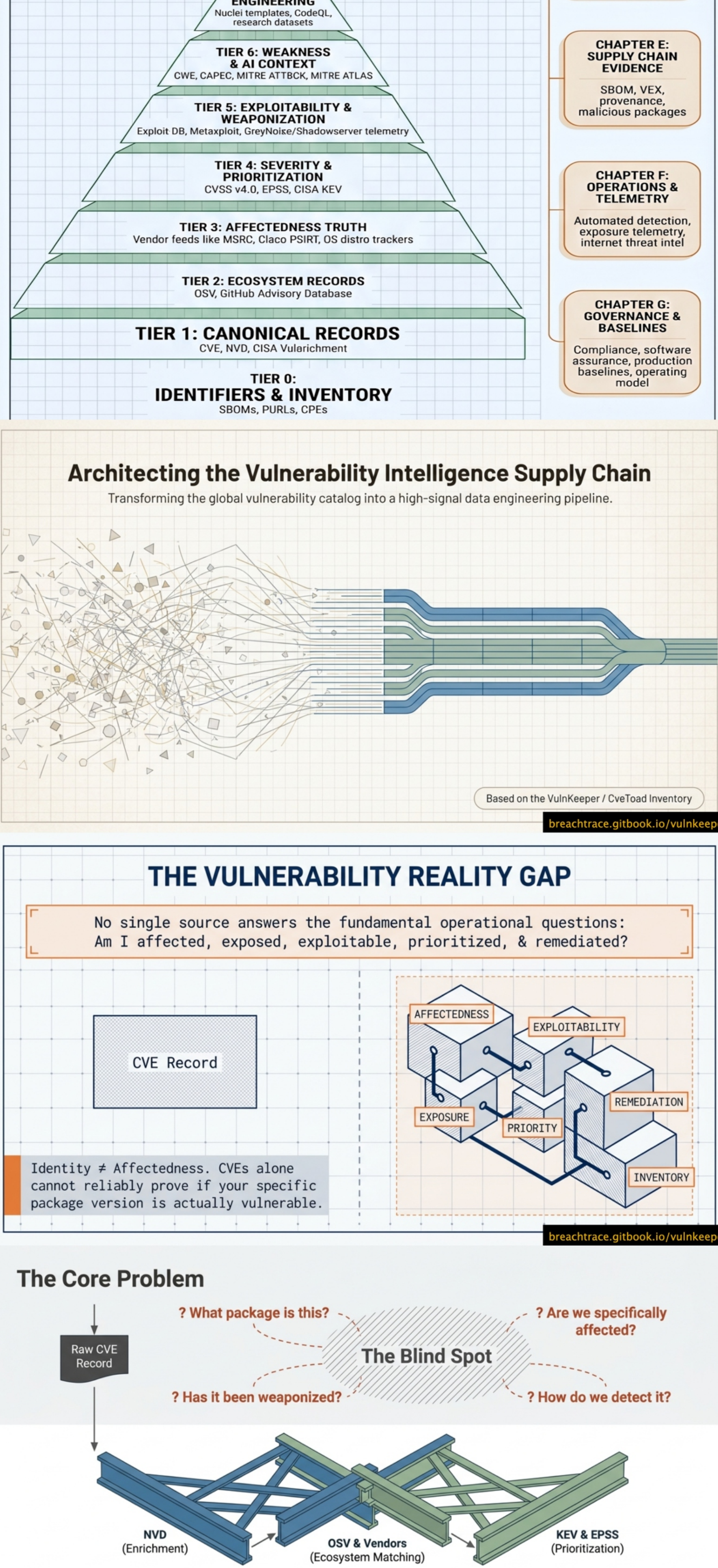
The Open Source Vulnerability Intelligence Refinery

Access the complete, continually updated source inventory and ingestion schemas.

breachtrace.gitbook.io/vulnkeeper
CveToad Vulnerability Management Source Inventory Repository

Licensed under GNU AGPL v3.

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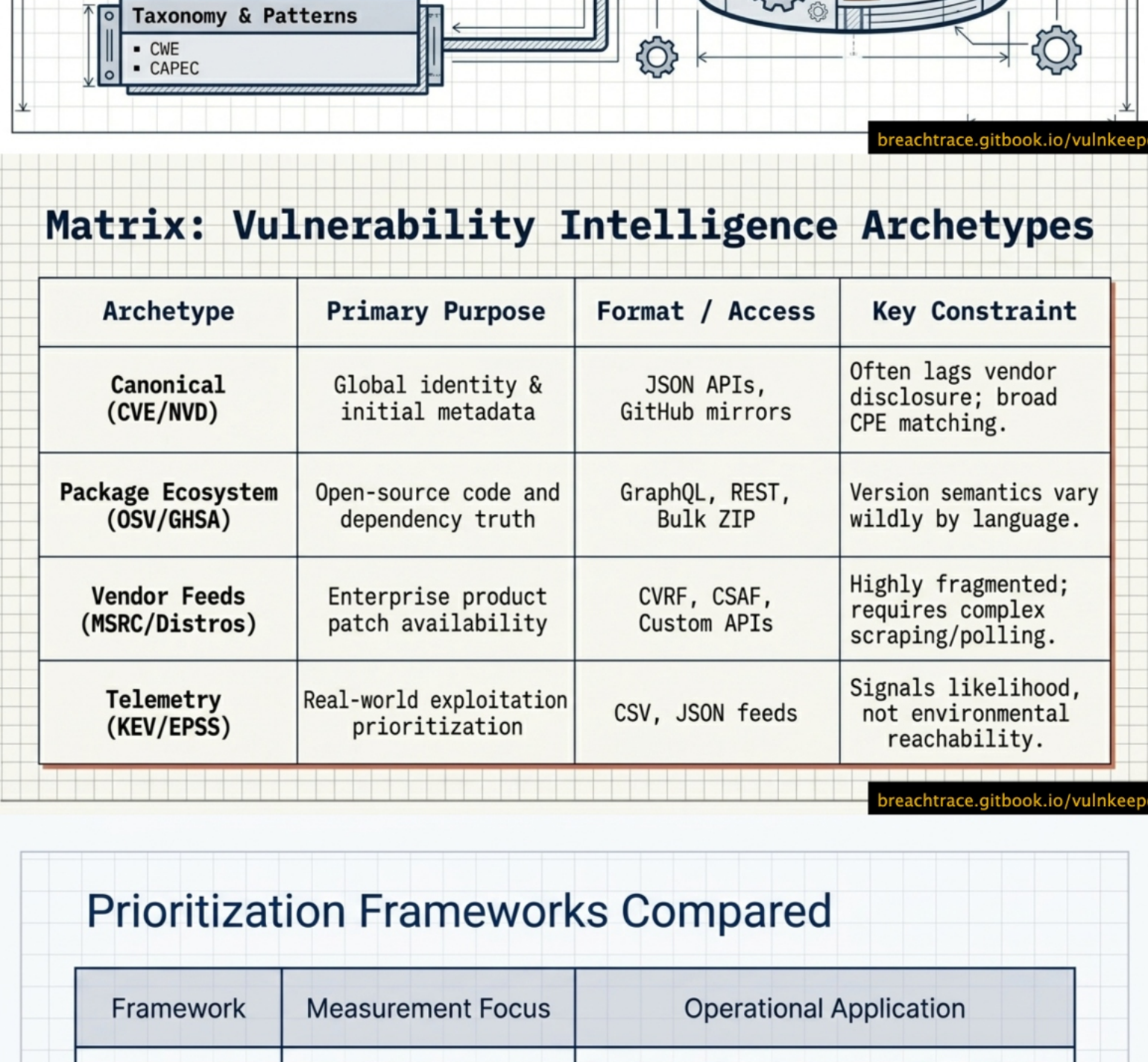
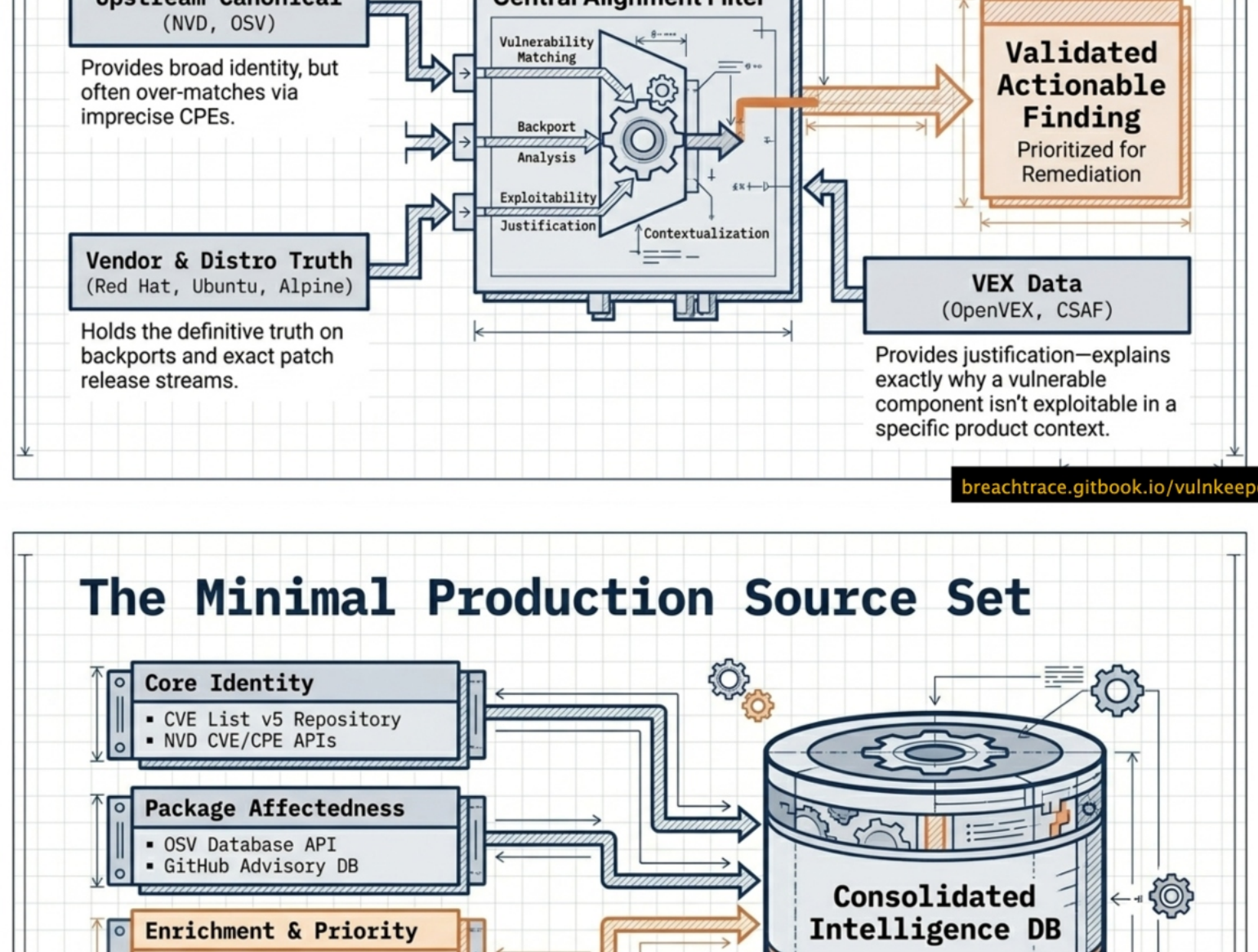
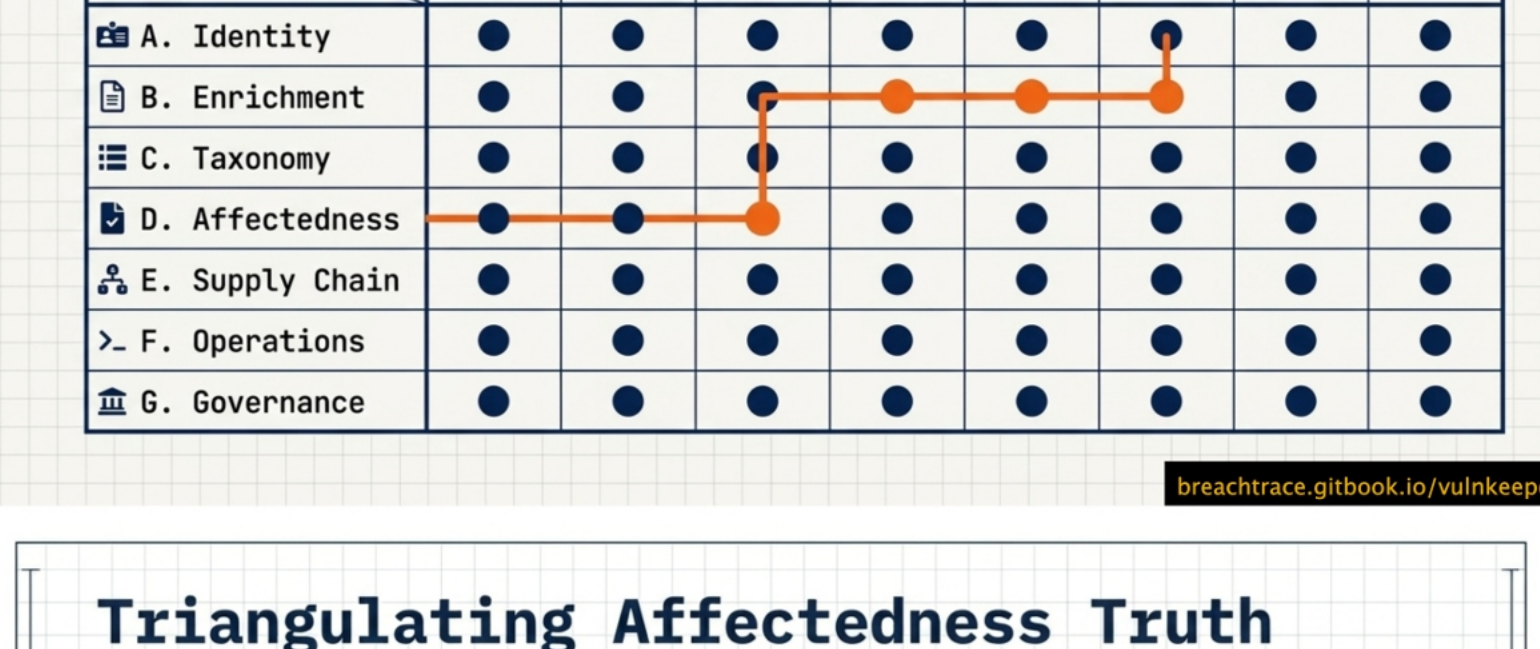
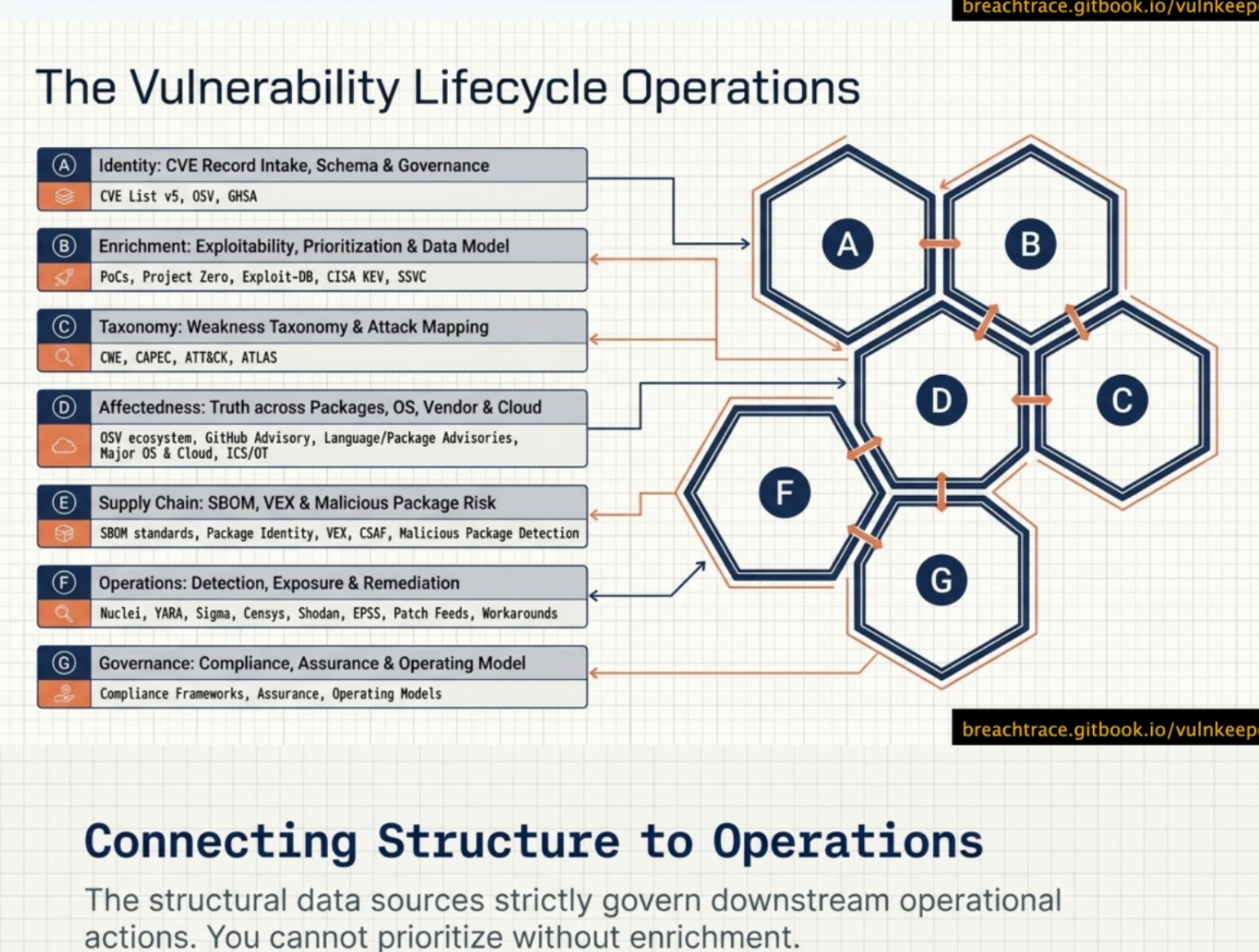
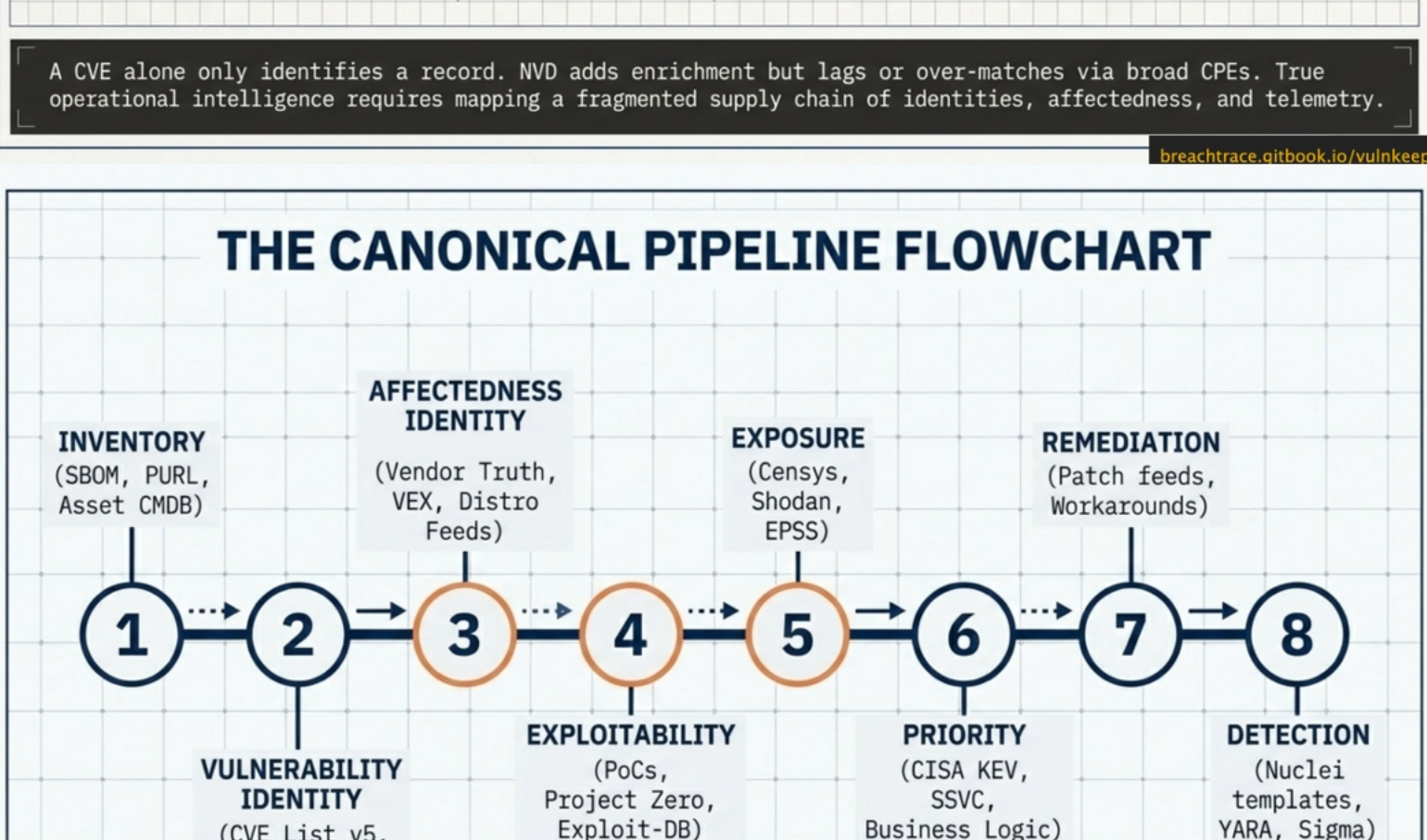


Architecting the Vulnerability Intelligence Supply Chain

Transforming the global vulnerability catalog into a high-signal data engineering pipeline.

Based on the VulnKeeper / CveToad Inventory

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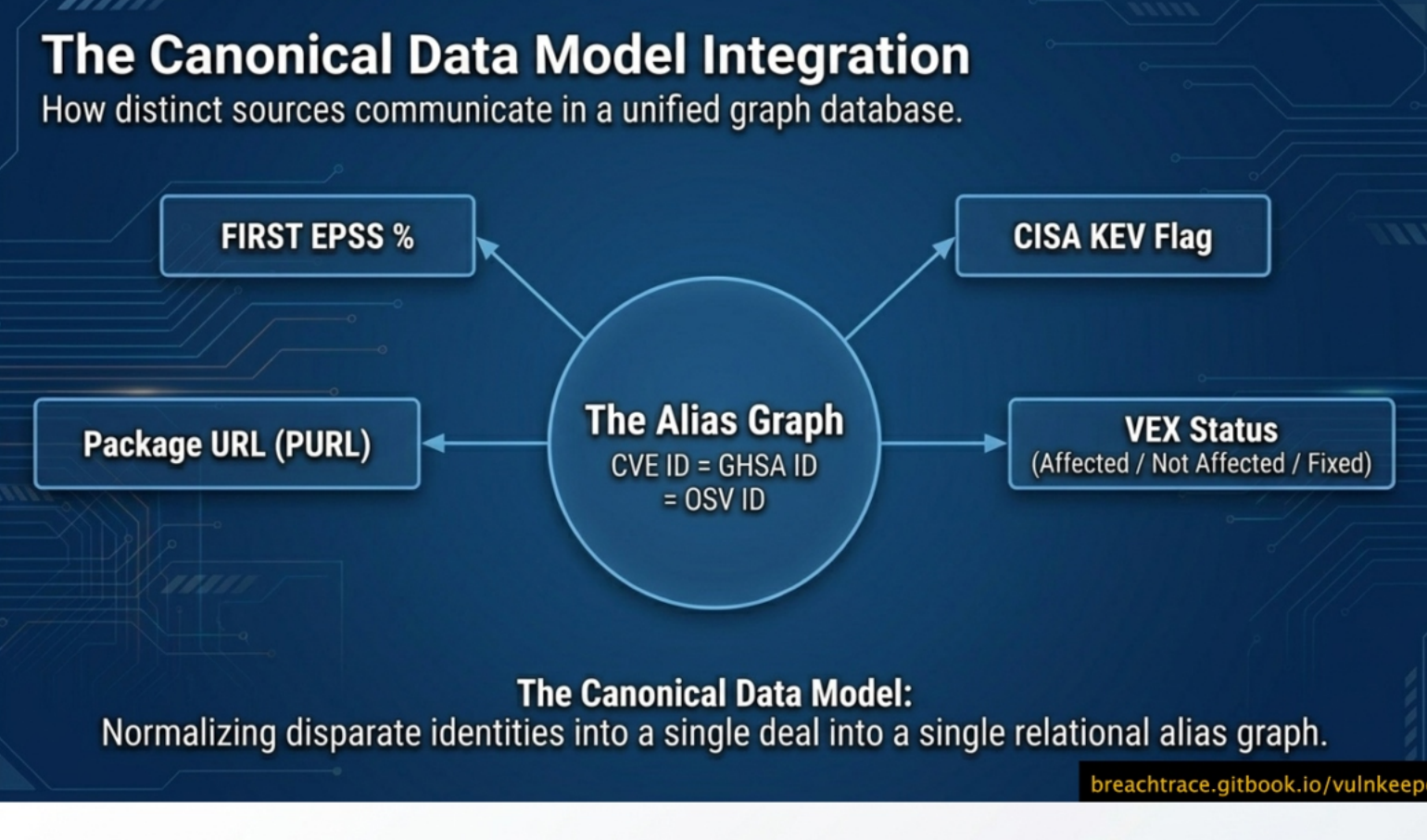
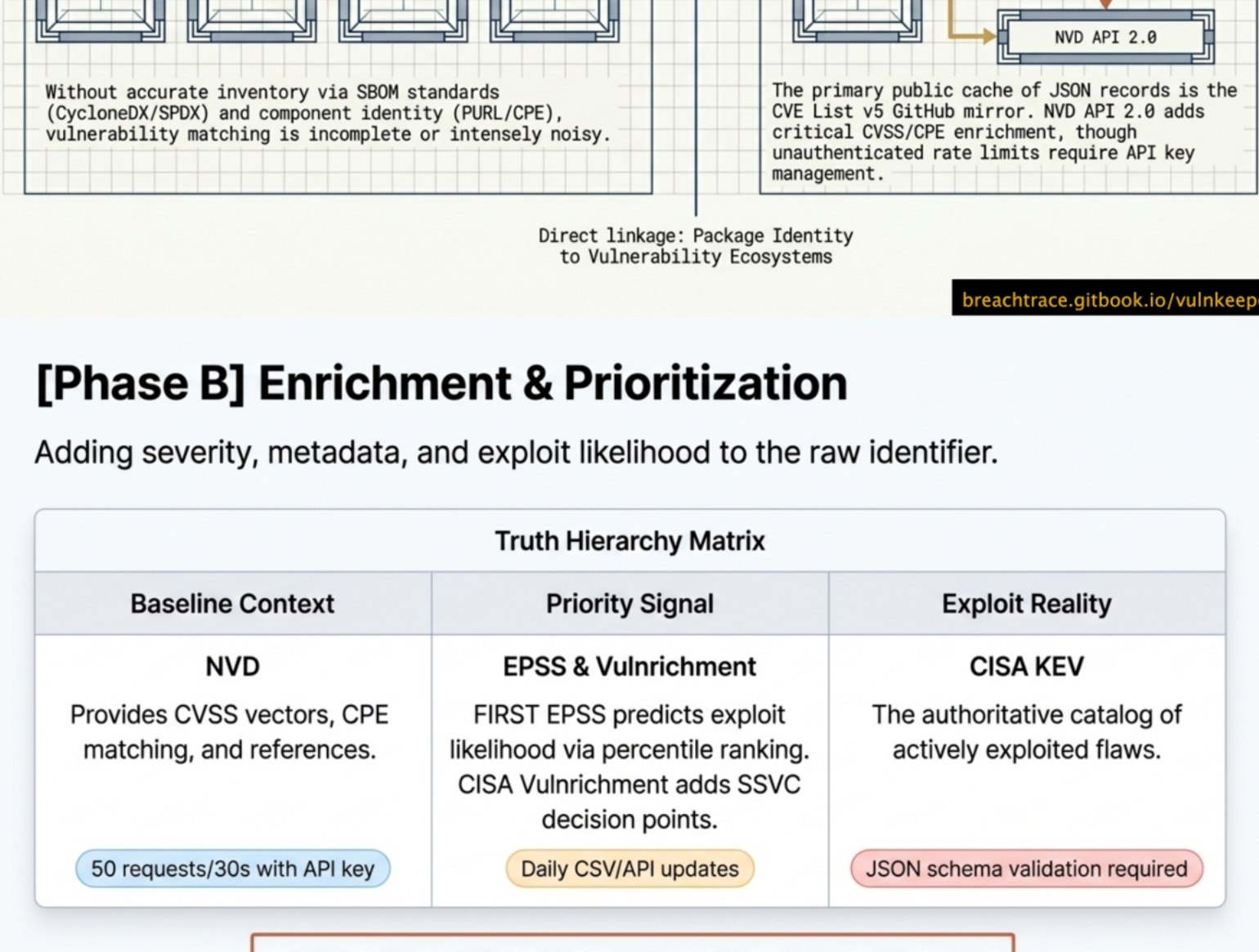


Connecting Structure to Operations

The structural data sources strictly govern downstream operational actions. You cannot prioritize without enrichment.

Y-Axis Chapter	X-Axis	1. Inventory	2. Identity	3. Affectedness	4. Exploitability	5. Exposure	6. Priority	7. Remediation	8. Detection
A. Identity		●	●	●	●	●	●	●	●
B. Enrichment		●	●	●	●	●	●	●	●
C. Taxonomy		●	●	●	●	●	●	●	●
D. Affectedness		●	●	●	●	●	●	●	●
E. Supply Chain		●	●	●	●	●	●	●	●
F. Operations		●	●	●	●	●	●	●	●
G. Governance		●	●	●	●	●	●	●	●

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Matrix: Vulnerability Intelligence Archetypes

Archetype	Primary Purpose	Foxmat / Access	Key Constraint
Canonical (CVE/NVD)	Global identity & initial metadata	JSON APIs, GitHub mirrors	Often lags vendor disclosure; broad CPE matching.
Package Ecosystem (OSV/GHSA)	Open-source code and dependency truth	GraphQL, REST, Bulk ZIP	Version semantics vary wildly by language.
Vendor Feeds (MSRC/Distro)	Enterprise product patch availability	CVRF, CSAF, Custom APIs	Highly fragmented; requires complex scraping/polling.
Telemetry (KEV/EPSS)	Real-world exploitation prioritization	CSV, JSON feeds	Signals likelihood, not environmental reachability.

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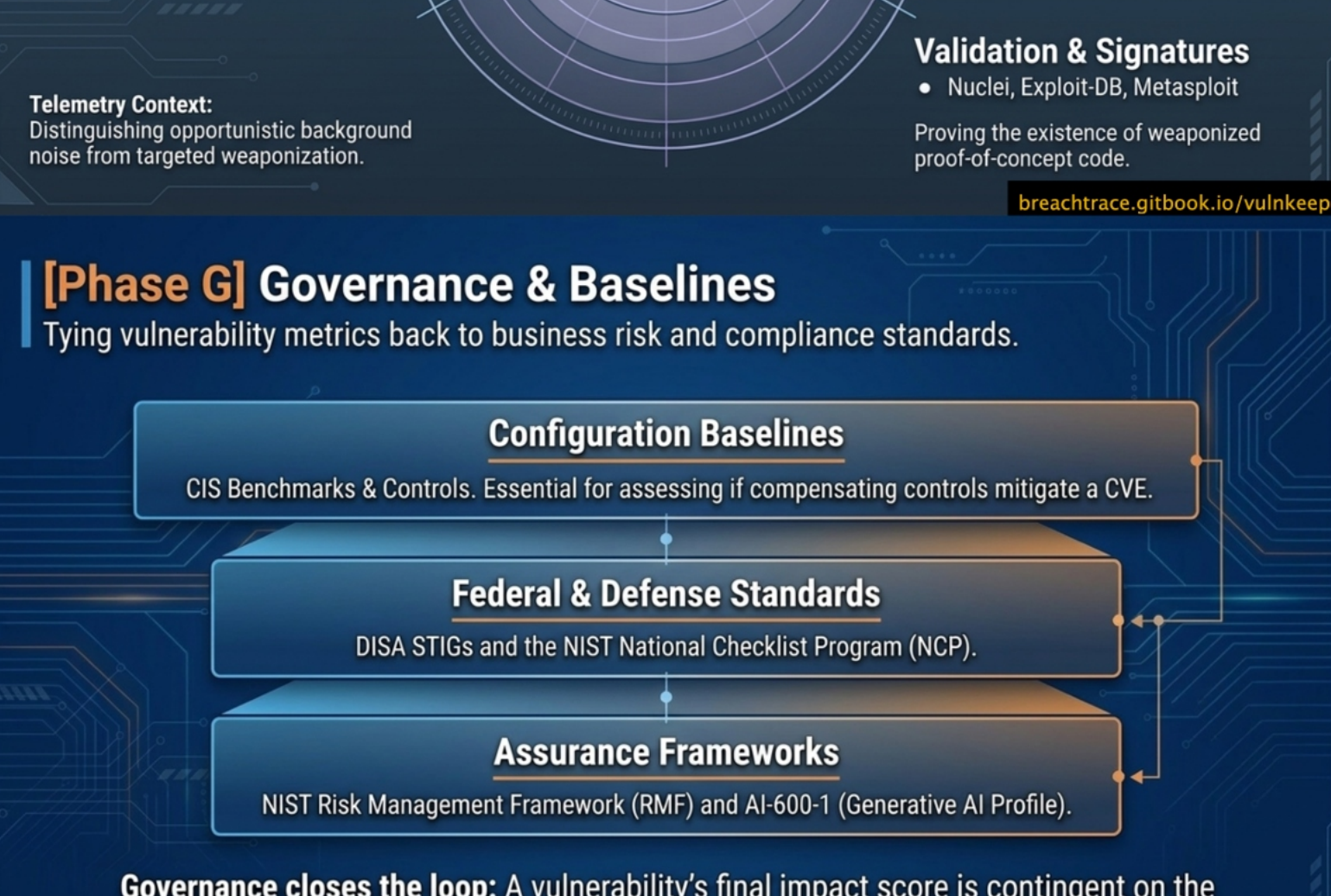
Prioritization Frameworks Compared

Framework	Measurement Focus	Operational Application
CVSS v4.0	Technical Severity	Baseline technical measurement. Warning: Do not use for exploit likelihood.
FIRST EPSS	Exploit Likelihood	Daily prediction of in-the-wild weaponization probability. Temporal and dynamic.
CISA KEV	Exploited Reality	Binary proof that a vulnerability is actively weaponized and exploited. Strongest operational signal.
CISA SSVC	Decision Logic	Tree-based model combining threat, asset context, and impact to mandate remediation timelines.

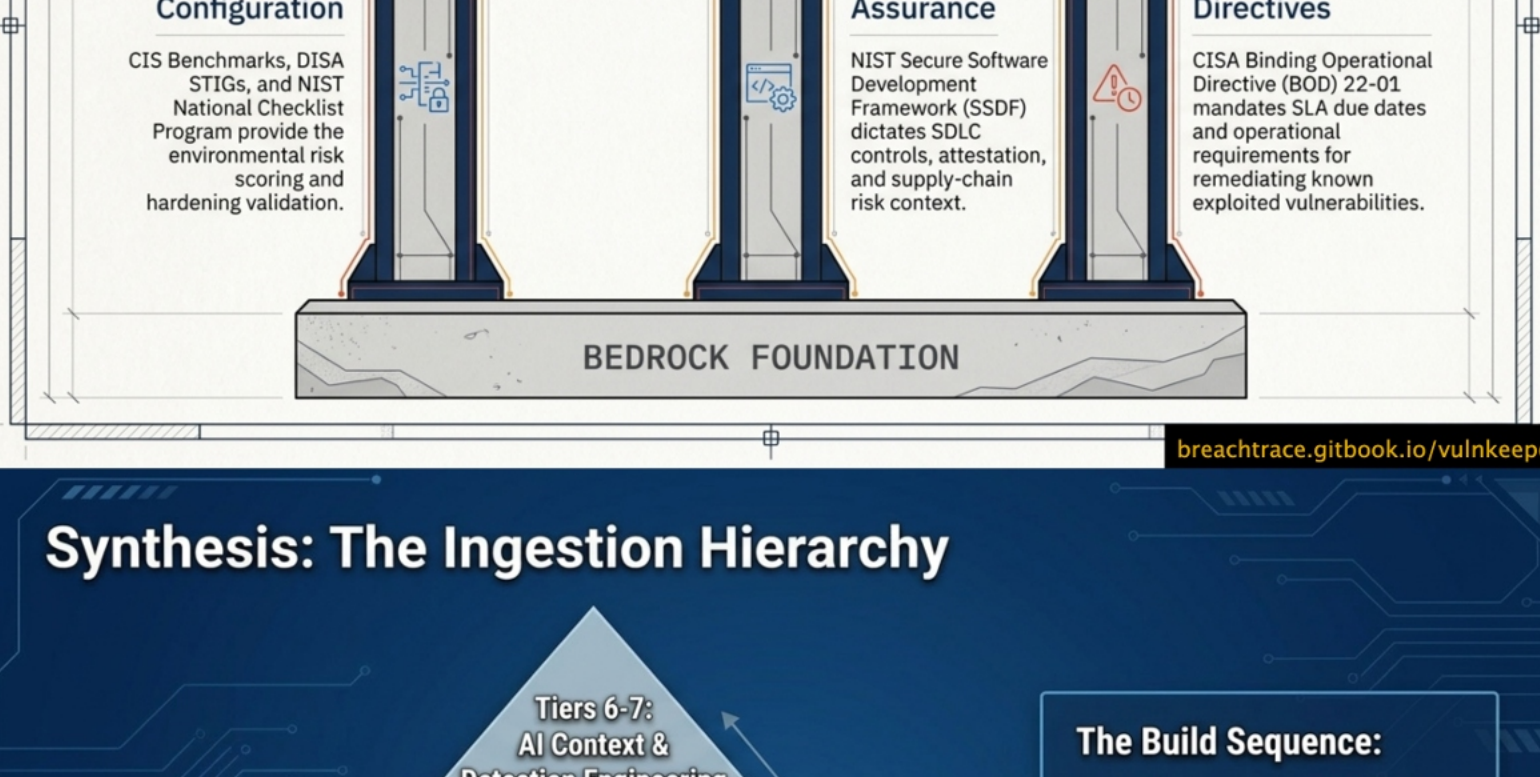
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[Phase A] Identity, Intake & Schema

Establishing the canonical root of truth. What vulnerability is this?

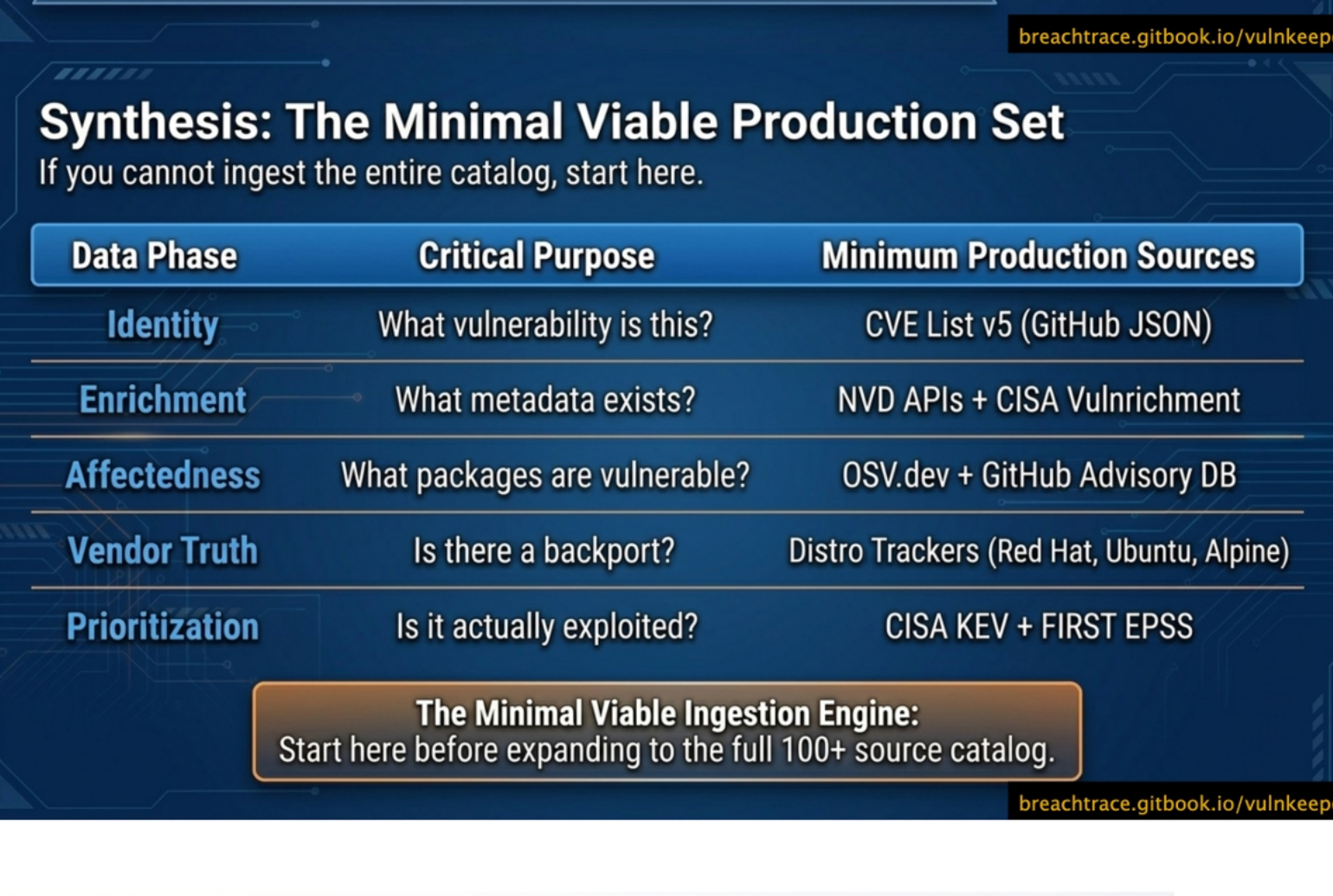


Tiers 0 & 1: You Cannot Patch What You Cannot See



[Phase B] Enrichment & Prioritization

Adding severity, metadata, and exploit likelihood to the raw identifier.



The Canonical Data Model Integration

How distinct sources communicate in a unified graph database.



[Phase C] Taxonomy & Attack Mapping

Transforming static CVE attributes into dynamic threat hunting queries using STIX/TAXII frameworks.

[Phase D] Affectedness Truth

[Phase E] Supply Chain Evidence

Binding vulnerabilities to actual internet inventory.

Steps 4 & 5: Severity ≠ Exploitability

[Phase F] Detection & Exposure

Validating if the vulnerability is externally reachable or actively targeted.

[Phase G] Governance & Baselines

Tying vulnerability metrics back to business risk and compliance standards.

The Bedrock: Governance, Assurance & Compliance

Synthesis: The Ingestion Hierarchy

Synthesis: The Minimal Viable Production Set

If you cannot ingest the entire catalog, start here.

Data Phase	Critical Purpose	Minimum Production Sources
Identity	What vulnerability is this?	CVE List v5 (GitHub JSON)
Enrichment	What metadata exists?	NVD APIs + CISA Vulnrichment
Affectedness	What packages are vulnerable?	OSV.dev + GitHub Advisory DB
Vendor Truth	Is there a backport?	Distro Trackers (Red Hat, Ubuntu, Alpine)
Prioritization	Is it actually exploited?	CISA KEV + FIRST EPSS

The Minimal Viable Ingestion Engine:
Start here before expanding to the full 100+ source catalog.

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The VulnKeeper Blueprint: Mapping the Vulnerability Intelligence Pipeline

Operational Imperatives

